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3.3 Derivative of trigonometric functions

sin cos - -

Derivatives of Trigonometric Functions

$$\rightarrow \frac{d}{dx}(\sin x) = \cos x \qquad \frac{d}{dx}(\csc x) = -\csc x \cot x$$

$$\frac{d}{dx}(\cos x) = -\sin x \qquad \frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x \qquad \frac{d}{dx}(\cot x) = -\csc^2 x$$

$$\sec(x) = \frac{1}{\cos(x)}$$

$$\cot(x) = \frac{1}{\tan(x)}$$

$$\csc(x) = \frac{1}{\sin(x)}$$

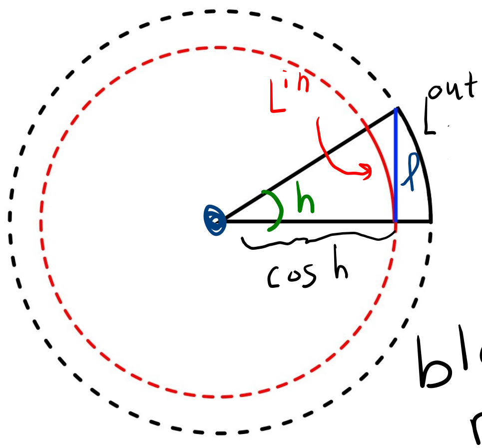
$$f(x) = \sin(x)$$

$$f'(x) = \lim_{h \to 0} \frac{\sin(x+h) - \sin(x)}{h}$$

$$= \lim_{h \to 0} \frac{\sin(x)\cos(h) + \sin(h)\cos(x) - \sin x}{h}$$

$$= \lim_{h \to 0} \frac{\sin(x)(\cos h - 1)}{h} + \lim_{h \to 0} \frac{\sin h \cos(x)}{h}$$

$$= \cos(x)$$



$\ell = \sin h$

$\sin^2 h + \cos^2 h = 1$

black circle
radius 1

$$L^{in} \le \ell \le L^{out}$$

$$h\cos h \le \sin h \le h$$

take h small

$$\cos h = \sqrt{1-\sin^2 h}$$

$$\frac{\cancel{h}\sqrt{1-\sin^2 h}}{\cancel{h}} \le \frac{\sin h}{h} \le \frac{\cancel{h}}{\cancel{h}} \quad 1$$

$$1 = \lim_{h\to 0}\sqrt{1-\sin^2 h} \le \lim_{h\to 0}\frac{\sin h}{h} \le \lim_{h\to 0} 1 = 1$$

Squeeze thm $\Rightarrow \boxed{\lim_{h\to 0}\frac{\sin h}{h} = 1}$

$$\sqrt{1-\sin^2 h} - 1 = \cos h - 1 \iff \left(\sqrt{1-\sin^2 h} - 1\right)\frac{\left(\sqrt{1-\sin^2 h}+1\right)}{\left(\sqrt{1-\sin^2 h}+1\right)} = \frac{\cancel{1}-\sin^2 h - \cancel{1}}{\sqrt{1-\sin^2 h}+1} = \cos h - 1$$

$$f(x) \le g(x) \le h(x) \Rightarrow \lim_{x\to a} g(x) = L$$

$$\lim_{x\to a} f(x) = \lim_{x\to a} h(x) = L$$

$$\lim_{h\to 0}\frac{\sin^2 h}{h\left(\sqrt{1-\sin^2 h}+1\right)} = \lim_{h\to 0}\frac{\cos h - 1}{h}$$

$$\boxed{0 = \lim_{h\to 0}\frac{\cos h - 1}{h}}$$

3.4 Chain rule

The Chain Rule If g is differentiable at x and f is differentiable at $g(x)$, then the composite function $F = f \circ g$ defined by $F(x) = f(g(x))$ is differentiable at x and F' is given by the product

$$\boxed{1} \qquad F'(x) = f'(g(x)) \cdot g'(x)$$

In Leibniz notation, if $y = f(u)$ and $u = g(x)$ are both differentiable functions, then

$$\boxed{2} \qquad \frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$$

Example 1) $\left(\underbrace{\sin}_{f}\left(\underbrace{\cos(x)}_{g}\right)\right)' = \cos(\cos(x)) \cdot -\sin(x)$

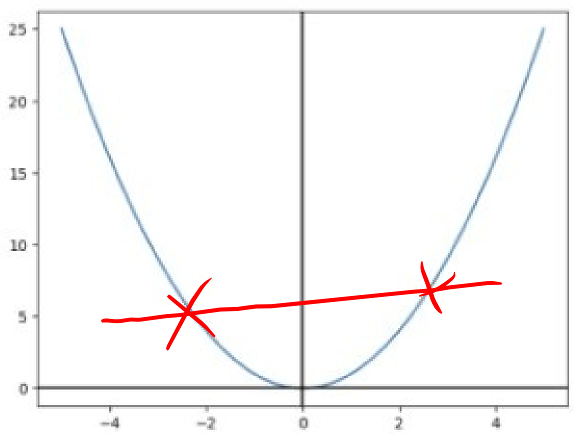
2) $\left(\underbrace{e}_{f}^{\underbrace{\sin(x)}_{g}}\right)' = e^{\sin(x)} \cos(x)$

3) $b^x = e^{x \ln b}$

$(b^x)' = \left(\underbrace{e}_{f}^{\underbrace{x \ln b}_{g}}\right)' = e^{x \ln b} \ln b$

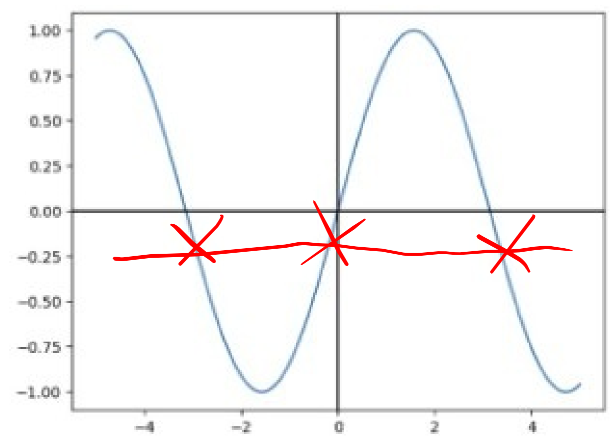
$= b^x \ln b$

A



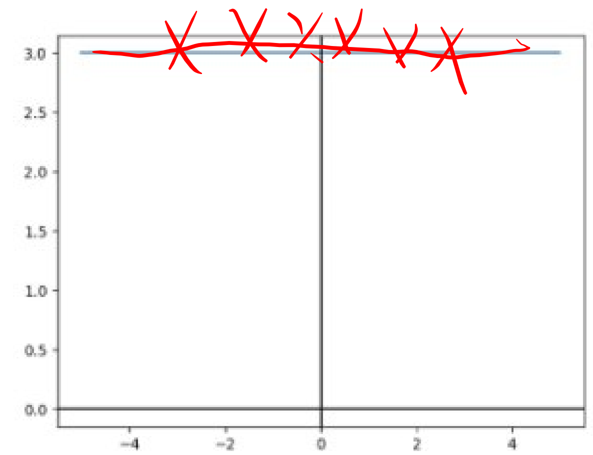
25
20
15
10
5
0
−4
−2
0
2
4

B



1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−4
−2
0
2
4

C



3.0
2.5
2.0
1.5
1.0
0.5
0.0
−4
−2
0
2
4